

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) What is the mean of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 10
- 2) What is the median of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 9
- 3) The six observations 4, 6, 6, 8, 10, 14 have mean 8 and squared-deviation sum 64. What is their population variance? (1)
(A) $64/6$
(B) $64/5$
(C) $\sqrt{64/6}$
(D) 8
- 4) A distribution has mean 8, median 7 and positive standard deviation. What is the sign of Pearson's second skewness coefficient? (1)
(A) Zero
(B) Negative
(C) Undefined
(D) Positive
- 5) Which measure of central tendency is generally less affected by an extreme observation? (1)
(A) Arithmetic mean
(B) Median
(C) Midrange
(D) All equally affected

Section B: Theory questions (25 marks)

- 6) Find the five-number summary and interquartile range of 3, 5, 7, 8, 9, 11, 13, 16, 30. Exclude the median when forming the lower and upper halves. Use the 1.5 *IQR* rule to identify outliers and sketch a box plot. (3)
- 7) A table contains customer ID, age, monthly income and annual income. Give one suitable treatment for missing ages and explain how you would detect inconsistency between the two income fields. (2)
- 8) A filling machine is claimed to dispense a mean of 500 ml. A random sample of 36 bottles has mean 496 ml. The population standard deviation is known to be 12 ml. Perform a two-sided *z*-test at the 5% level. State the hypotheses, statistic, decision and conclusion. Use critical values ± 1.96 . (5)
- 9) Two judges rank six projects. Judge A assigns ranks (1, 2, 3, 4, 5, 6) and judge B assigns (2, 1, 4, 3, 6, 5). Calculate Spearman's rank correlation coefficient and interpret the result. (3)
- 10) Partition 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40 into three equal-frequency bins. Replace each observation by its bin mean and report the three resulting bin means. (2)
- 11) Test independence between training and passing an assessment using Pearson's chi-square test at 5%. Of 60 trained students, 48 pass and 12 fail; of 40 untrained students, 22 pass and 18 fail. Show expected frequencies and the statistic. Use $\chi^2_{0.05,1} = 3.841$; do not apply a continuity correction. (5)
- 12) Three independent normal populations have equal variances. Samples are A: 4, 5, 6, B: 7, 8, 9, C: 10, 11, 12. The between-group sum of squares is 54 and within-group sum of squares is 6. Complete the one-way ANOVA degrees of freedom, mean squares and *F* statistic. Use $F_{0.05;(2,6)} = 5.143$ and state your conclusion. (3)
- 13) Distinguish a Type I error from a Type II error in the context of the filling-machine test in Question 8. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which C keyword defines a structure type? (1)
(A) `class`
(B) `union`
(C) `struct`
(D) `enum`
- 2) Object `s` has a nested structure member `admitted` with an integer member `year`. How do you access that `year`? (1)
(A) `s.admitted.year`
(B) `s->admitted.year`
(C) `s.year.admitted`
(D) `s::admitted::year`
- 3) How many non-null characters can a null-terminated string in `char name[40]` hold? (1)
(A) 40
(B) 41
(C) 38
(D) 39
- 4) Which `scanf` format limits a single word to fit in `char name[40]`, including its null terminator? (1)
(A) `%40s`
(B) `%39s`
(C) `%s`
(D) `%41s`
- 5) If `p` points to a structure with integer member `roll`, which expression accesses that member? (1)
(A) `p.roll`
(B) `*p.roll`
(C) `p->roll`
(D) `p::roll`

Section B: Theory questions (25 marks)

- 6) Write a C function that dynamically allocates a contiguous $r \times c$ integer matrix and returns its base pointer, or NULL on failure. Reject nonpositive dimensions. Show how element (i, j) is accessed using pointer arithmetic. Assume the requested size fits in `size_t`. (3)
- 7) Explain a dangling pointer and a memory leak using one short C example of each. State how each error can be prevented. (2)
- 8) A singly linked list stores integer values. Define its node structure and write `void deleteAll(Node **head, int key)` to delete every node whose value equals `key`. Handle an empty list, consecutive matches and matches at the head. Release deleted nodes. (5)
- 9) Write a C function to reverse a singly linked list in place using three pointers. Return the new head and state its time and auxiliary-space complexity. (3)
- 10) A list initially contains $10 \rightarrow 20 \rightarrow 30$. Insert 5 at the head, insert 25 after 20, then delete the tail. Draw the final list and state how many next links must be traversed to locate its tail from the head. (2)
- 11) Convert $A+B*(C-D)^E^F$ to postfix using a stack. Exponentiation has highest precedence and is right-associative. Show the operator stack and output after each input token, then give the final postfix expression. (5)
- 12) Write a C function to check whether a string containing only `(, ), [, and ]` is balanced. Use an array stack of capacity 100 and assume the input has at most 100 characters. Return 1 for balanced and 0 otherwise. (3)
- 13) An editor uses undo and redo stacks. Starting with empty stacks, perform edits A, B and C, then undo twice and redo once. List both stacks from bottom to top. State what happens to the redo stack when a new edit D is made. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A noiseless channel has bandwidth 3 kHz and four signal levels. What is its Nyquist maximum bit rate? (1)
(A) 3 kbps
(B) 6 kbps
(C) 12 kbps
(D) 24 kbps
- 2) How many bits can one of four distinct signal levels represent? (1)
(A) 1
(B) 2
(C) 4
(D) 8
- 3) A channel has bandwidth 3 kHz and linear signal-to-noise ratio 31. What is its Shannon capacity? (1)
(A) 15 kbps
(B) 12 kbps
(C) 93 kbps
(D) 30 kbps
- 4) For the same channel and signalling scheme, Nyquist gives 12 kbps and Shannon gives 15 kbps. Which is the tighter upper bound? (1)
(A) 15 kbps
(B) 27 kbps
(C) 3 kbps
(D) 12 kbps
- 5) Under the noiseless Nyquist model, what happens to the maximum bit rate when bandwidth doubles and the number of levels stays fixed? (1)
(A) It halves
(B) It stays fixed
(C) It doubles
(D) It quadruples

Section B: Theory questions (25 marks)

- 6) Compare ASK, binary FSK and BPSK in terms of the carrier parameter changed and sensitivity to amplitude noise. State one reason to prefer BPSK over ASK in a channel with amplitude disturbances. (3)
- 7) An application generates 900 payload bytes. Each of five layers adds a 20-byte header. Calculate the total transmitted size and the percentage occupied by headers, ignoring trailers. (2)
- 8) Encode data 1101011011 using CRC generator 10011. Show modulo-2 division, the four-bit remainder and the complete transmitted codeword. State the receiver's check for an error-free received word. (5)
- 9) Draw Manchester and bipolar AMI waveforms for 10110010. For Manchester, use low-to-high at mid-bit for 1 and high-to-low for 0. For AMI, assume the last nonzero pulse before this sequence was negative. Label all bit boundaries. (3)
- 10) Explain the different actions of a hub and a learning switch when a frame arrives. For the switch, distinguish known and unknown destination MAC addresses. (2)
- 11) A stop-and-wait link sends 1000-byte data frames at 1 Mbps. One-way propagation delay is 10 ms. ACK transmission and processing times are negligible; no frames are lost. Calculate frame transmission time, time per successful cycle, utilization and useful throughput. Ignore headers. (5)
- 12) A selective-repeat sender sends frames 0, 1, 2 and 3 with window size 4. Frame 1 is lost; frames 0, 2 and 3 arrive and their individual ACKs return before timeout. No new frames are available. State which frames are buffered, which frame is retransmitted and the final in-order delivery sequence. Compare with go-back-N under the same loss, assuming it discards out-of-order frames. (3)
- 13) A shared half-duplex Ethernet link operates at 100 Mbps. The maximum one-way propagation delay is $5\ \mu\text{s}$. Ignoring other delays, calculate the minimum frame length required for collision detection in bits and bytes. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) How many cells are in a four-variable Karnaugh map? (1)
(A) 4
(B) 8
(C) 16
(D) 32
- 2) How many input variables differ between two adjacent Karnaugh-map cells? (1)
(A) 1
(B) 2
(C) 3
(D) 4
- 3) Which is a valid group size when minimizing a sum-of-products expression on a Karnaugh map? (1)
(A) 3
(B) 5
(C) 6
(D) 8
- 4) For $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$, with A most significant, which expression is equivalent? (1)
(A) $B \oplus D$
(B) $\neg B \neg D + BD$
(C) $B + D$
(D) BD
- 5) How can a two-input NAND gate act as an inverter for input X? (1)
(A) Tie both inputs to 0
(B) Leave one input open
(C) Tie both inputs to X
(D) Tie one input to 0

Section B: Theory questions (25 marks)

- 6) Using 5-bit two's complement, evaluate (i) $11 + 7$, (ii) $-9 + 5$, and (iii) $6 - 13$. Give the binary result, its signed interpretation and whether signed overflow occurs in each case. (3)
- 7) Implement $F(A, B, C) = A \oplus B \oplus C$ using a 4-to-1 multiplexer. Choose A and B as select inputs and specify I_0, I_1, I_2, I_3 ; an inverter is available. (2)
- 8) Design a synchronous two-bit counter with T flip-flops and enable input E. For $E = 1$, the state sequence is $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$; for $E = 0$, the state holds. Draw the transition table, derive both T-input equations and draw the circuit. (5)
- 9) Write synthesizable Verilog for a four-bit ring counter with positive-edge clock and active-high synchronous reset. Reset must load 0001; subsequent clocks rotate the one toward the next more significant bit, wrapping from bit 3 to bit 0. (3)
- 10) Explain why an asynchronous ripple counter can briefly produce incorrect decoded outputs. Contrast its clock connection with that of a synchronous counter. (2)
- 11) Design a two-bit unsigned magnitude comparator for $A = a_1a_0$ and $B = b_1b_0$. Derive Boolean expressions for $A > B$, $A = B$ and $A < B$. Draw a gate-level diagram and verify the outputs for $A = 10_2$, $B = 01_2$. (5)
- 12) A byte-addressable memory stores 32-bit words. How many words fit in 4 KiB? How many address bits select individual bytes? If a word starts at hexadecimal address $0x1000$, list the byte addresses it occupies and the starting address of the next word. (3)
- 13) Describe a memory-read operation using the memory address register (MAR), memory data register (MDR) and read control signal. State what changes for a memory-write operation. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the least element? (1)
(A) 0
(B) 2
(C) 1
(D) 12
- 2) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the greatest element? (1)
(A) 1
(B) 12
(C) 6
(D) 4
- 3) In the divisor lattice D_{12} , what is $4 \wedge 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 4) In the divisor lattice D_{12} , what is $4 \vee 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 5) Which is a complement of 2 in D_{12} , with meet given by gcd and join by lcm? (1)
(A) 3
(B) 6
(C) No element
(D) 12

Section B: Theory questions (25 marks)

- 6) For $F(x, y, z) = (x \wedge \neg y) \vee z$, construct the truth table and write the canonical disjunctive normal form and canonical conjunctive normal form. Use variable order (x, y, z) . (3)
- 7) Prove the Boolean identity $a \vee (\neg a \wedge b) = a \vee b$ using distributive and complement laws. Name the laws used. (2)
- 8) Count the integer solutions of $x + y + z = 12$ subject to $1 \leq x \leq 5$, $2 \leq y \leq 6$ and $z \geq 0$. Transform to nonnegative variables and use inclusion-exclusion. Show each excluded case. (5)
- 9) Find the 42nd lexicographic permutation of the symbols 0, 1, 2, 3, 4, with 01234 counted as the first. Show the factorial-block choices. (3)
- 10) How many distinct arrangements of the letters of BALLOON are possible? How many have the two Ls adjacent? Explain how repeated letters are handled. (2)
- 11) Use Dijkstra's algorithm from A in an undirected weighted graph with vertices A, B, C, D, E and edges AB:4, AC:1, BC:2, BD:1, CD:5, CE:8, DE:3. Show the tentative-distance table after settling each vertex. Give a shortest path and distance from A to every other vertex. (5)
- 12) Prove that a simple graph with $n \geq 2$ vertices and minimum degree at least $(n - 1)/2$ is connected. Hint: bound the size of each component if the graph were disconnected. (3)
- 13) A simple graph is regular and self-complementary on n vertices. Use degree counting to show its degree is $(n - 1)/2$ and then show that $n \equiv 1 \pmod{4}$. (2)

End of question paper